INDIAN INSTITUTION OF TECHNOLOGY MANDI

AI APP COMPILER

INTELLRELIABILITY-FOCUSED APPLICATION GENERATION FRAMEWORK

P R E S E N T E D B Y - J A N H A V I H A T E K A R

PROBLEM STATEMENT

Current AI code generation systems can generate software artifacts quickly, but often suffer from:

- Missing requirements
- Inconsistent architectures
- Invalid schemas
- Lack of reliability checks
- No execution verification

There is a need for a system that not only generates applications but also validates, repairs, evaluates, and verifies them automatically.

PROPOSED SOLUTION

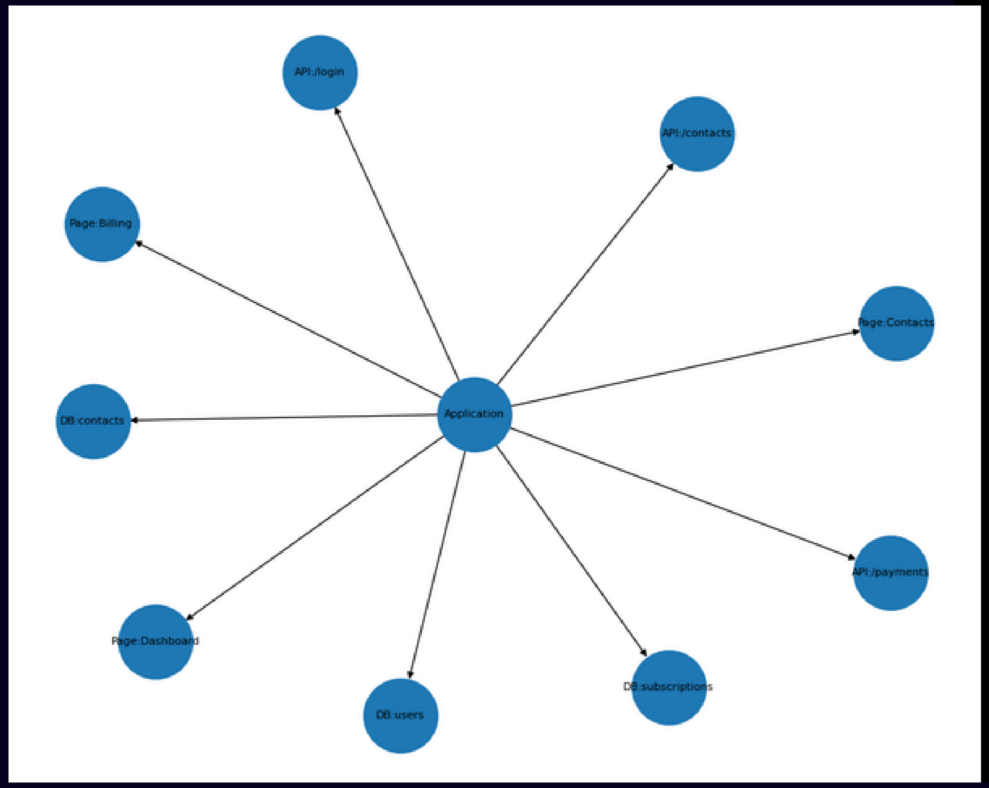
Proposed Solution: **AI App Compiler**

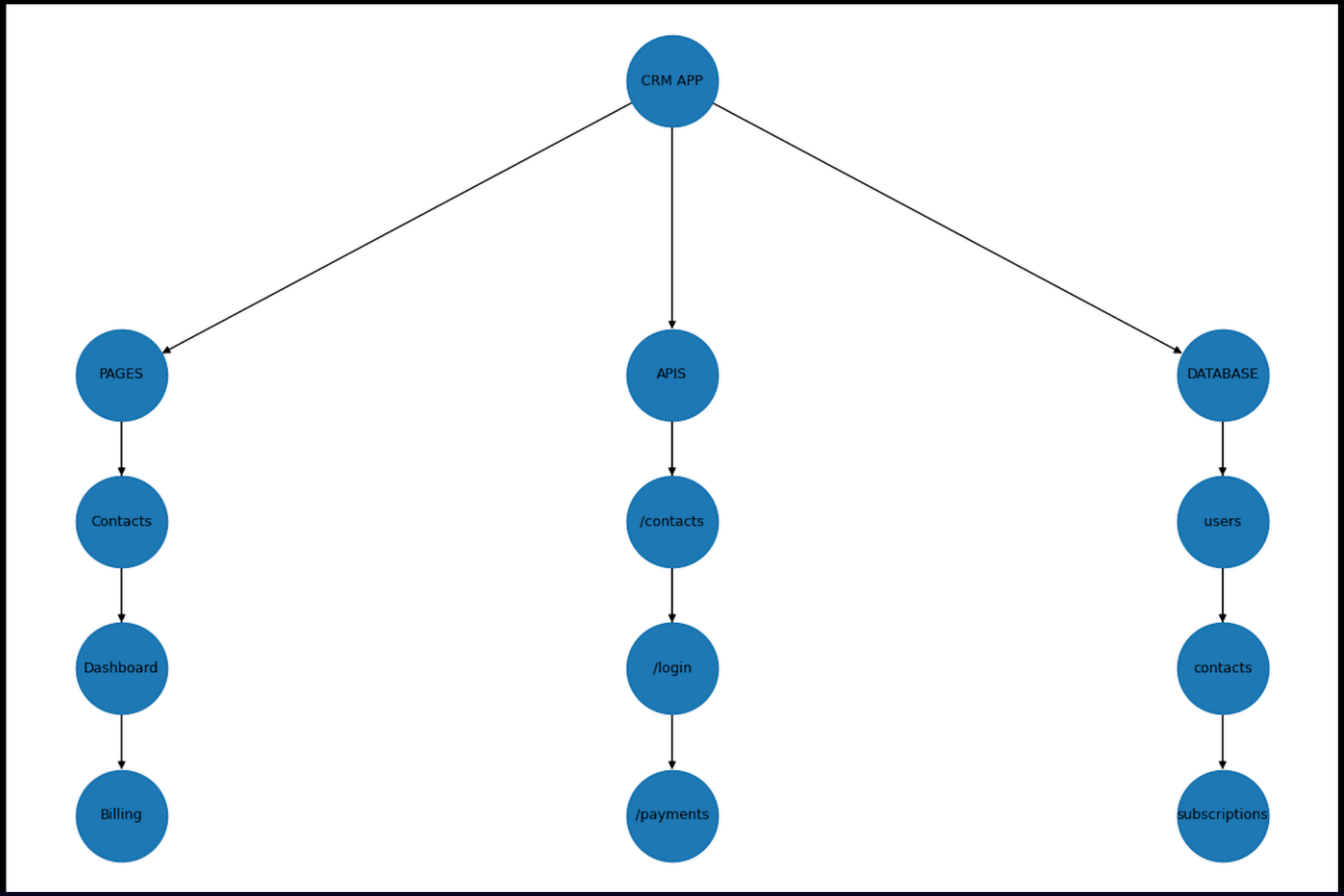
AI App Compiler transforms natural language requirements into validated application architectures and executable project artifacts.

Key Capabilities:
- Requirement Analysis
- Intent Extraction
- Architecture Planning
- Schema Generation
- Validation & Repair
- Reliability Scoring
- Runtime Generation
- Execution Validation
- Project Export

SYSTEM ARCHITECTURE

User Prompt
↓
Requirement Analyzer
↓
Clarification Engine
↓
Intent Extractor
↓
Architecture Planner
↓
Schema Generator
↓
Validation Engine
↓
Repair Engine
↓
Consistency Validator
↓
Reliability Score
↓
Runtime Generator
↓
Execution Validator
↓
Metrics Tracker
↓
ZIP Export





CORE COMPONMENT

01 Requirement Analyzer
Detects incomplete or ambiguous requirements.

02 Clarification Engine
Generates clarification questions.

03 Intent Extractor
Extracts application type, features, and roles.

04 Architecture Planner
Generates entities, pages, APIs, and database tables.

05 Schema Generator
Produces UI, API, Database, and Authentication schemas.

06 Validation & Repair Engine
Detects and fixes inconsistencies automatically.

RELIABILITY AND EVALUATION

The system evaluates generated outputs using:

- Validation Error Count
- Consistency Issues
- Repair Operations
- Execution Verification
- Reliability Score

Results:

Reliability Score : 100/100
Validation Errors : 0
Consistency Issues : 0
Execution Validation : PASS

RUNTIME GENERATION & API

Generated Runtime Artifacts

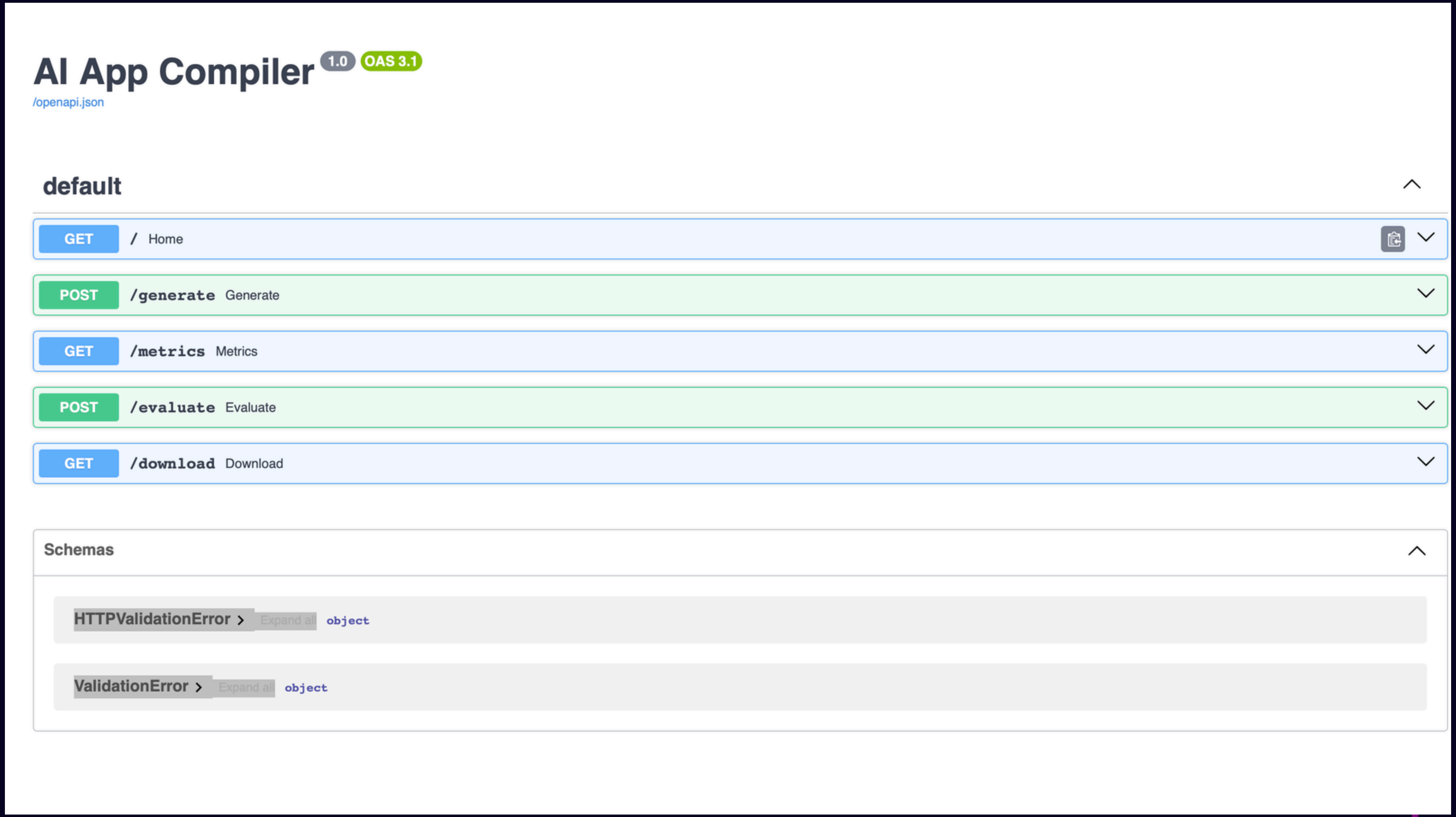
Generated Files:

- main.py
- models.py
- routes.py
- architecture_graph.png
- generated_app.zip

FastAPI Endpoints:

GET /
POST /generate
GET /metrics
POST /evaluate
GET /download

These endpoints expose the compiler as a reusable service.



DEMO RESULTS

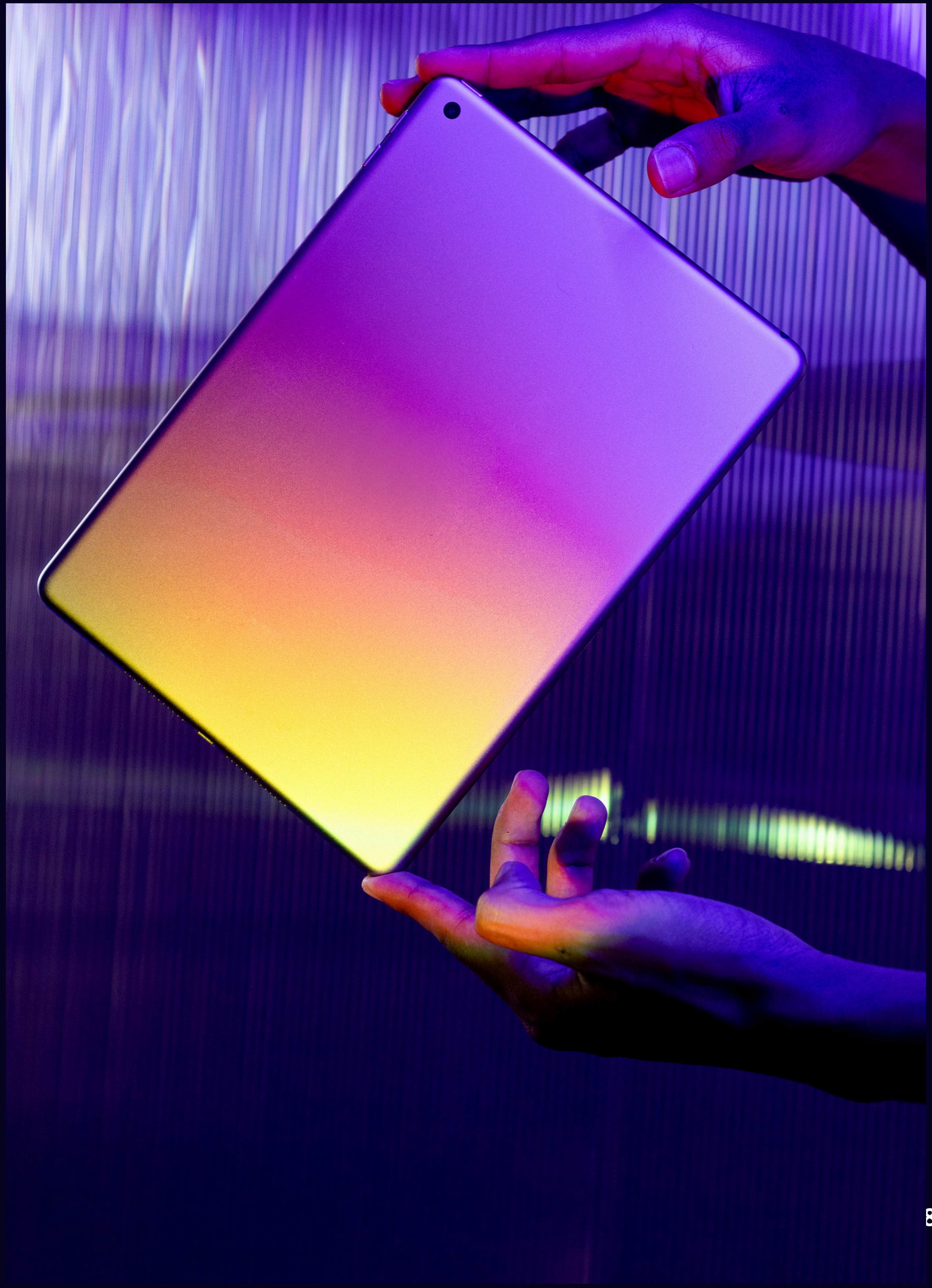
Demo Scenario

Prompt:

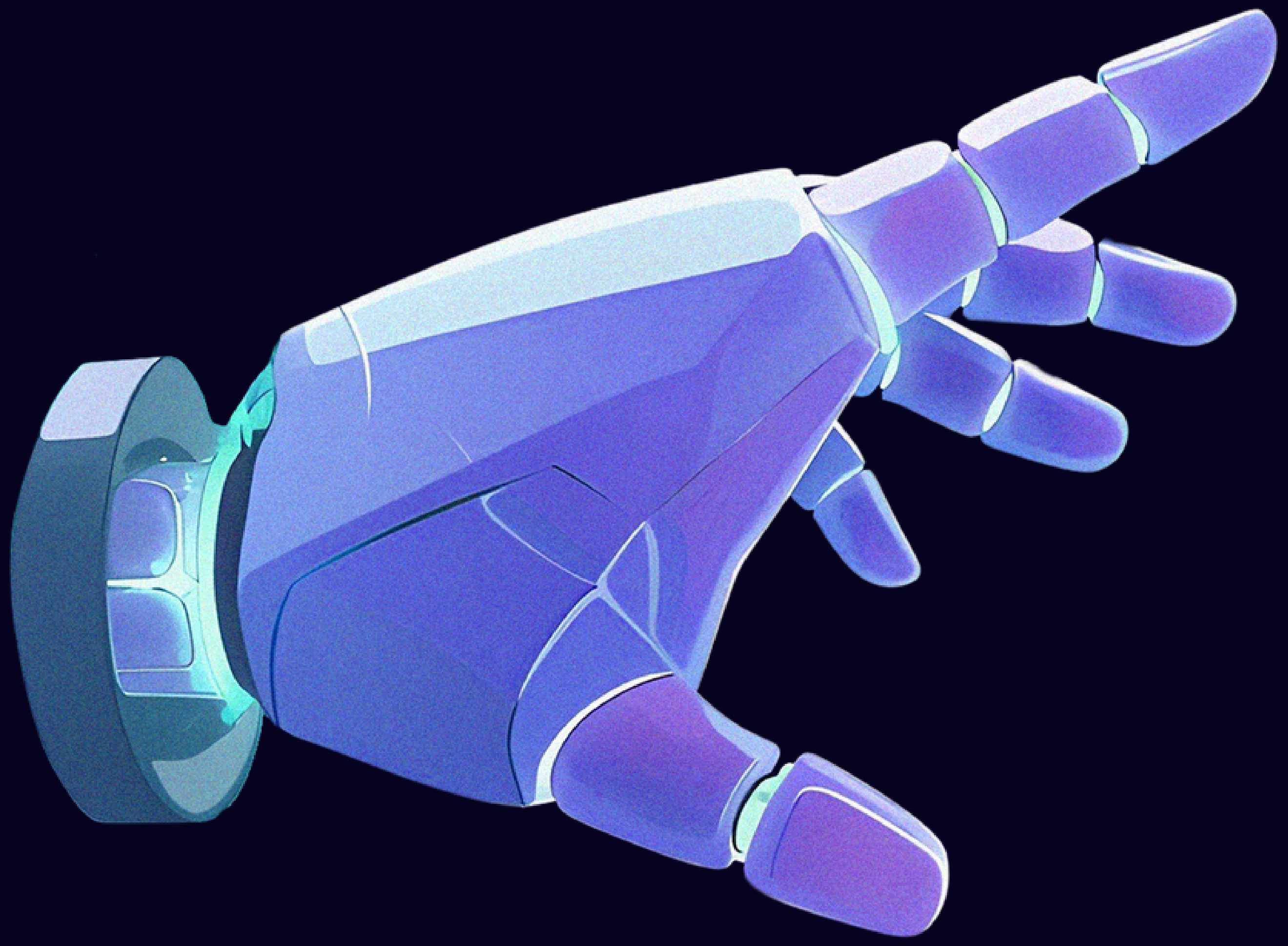
"Build CRM with login contacts dashboard analytics"

Generated Outputs:
- Application Architecture
- UI Schema
- API Schema
- Database Schema
- Authentication Schema
- Architecture Graph
- Executable Project Files

Pipeline executed successfully with Reliability Score = 100.



FUTURE WORK



Future Enhancements

- LLM-powered Intent Extraction
- Frontend Code Generation
- Docker Export
- Cloud Deployment
- Multi-language Support
- Advanced Runtime Generation
- CI/CD Integration
- Automated Testing Pipelines

CONCLUSION

AI App Compiler is a reliability-focused software generation framework that transforms natural language requirements into validated architectures, executable projects, and downloadable software packages.
The system improves trustworthiness in AI-generated applications through validation, repair, evaluation, and execution verification.

THANK YOU

github link : https://github.com/janhavi-5002/ai-app-compiler